

Olympics Data Analysis Report

Project Overview

This project analyzes historical Olympic athlete data to uncover participation trends, medal distributions, country performance, and athlete demographics. The goal is to transform raw Olympic data into meaningful insights using Python, PostgreSQL, SQL, and Power BI.

2. Dataset Summary

Dataset Information

- Rows: 70,000+
- Columns: 15
- Dataset Type: Historical Olympic Athlete Records

Key Features

- Athlete Information (Name, Sex, Age, Height, Weight)
- Participation Details- (Sport, Event, Season, Year)
- Country Information (NOC, Team) • Performance Metrics (Medal)

Missing Data Found

Age: 2732

Height: 16254

Weight: 17101

Medal: 60310

3. Exploratory Data Analysis using Python

The dataset was cleaned and prepared using Pandas.

Data Preparation

- Loaded dataset using Pandas.
- Examined dataset structure using info().

- Generated descriptive statistics using describe().

Missing Value Handling

- Age → Median Imputation
- Height → Median Imputation
- Weight → Median Imputation
- Medal → Replaced with 'No Medal'

Data Validation

All columns contained zero missing values after preprocessing.

Database Integration

The cleaned DataFrame was loaded into PostgreSQL using SQLAlchemy and Psycopg2.

Database: dataset_olympics

Table: Olympic

4. Data Analysis using SQL

The following analytical queries were performed:

1. Total Athletes

	total_athletes  bigint
1	70000

2. Male vs Female Participation

	Sex  text	athlete_count  bigint
1	M	51877
2	F	18123

3. Top 10 Sports by Participation

	Sport text	athlete_count bigint
1	Athletics	10629
2	Gymnastics	6248
3	Swimming	5964
4	Fencing	3096
5	Shooting	3040
6	Cycling	2794
7	Rowing	2690
8	Alpine Skiing	2326
9	Cross Country Skii...	2153
10	Football	1959

4. Top 10 Countries by Medal Count

	NOC text	medal_count bigint
1	USA	1561
2	FRA	660
3	GBR	604
4	ITA	604
5	URS	515
6	GER	434
7	AUS	373
8	SWE	365
9	CAN	360
10	NED	272

5. Participation Trend Over Years

	Year bigint 	participants bigint 
1	1896	65
2	1900	631
3	1904	308
4	1906	431
5	1908	865
6	1912	1025
7	1920	1218
8	1924	1579
9	1928	1434
10	1932	914

6. Medal Distribution

	Medal text 	total bigint 
1	No Med...	60310
2	Gold	3292
3	Bronze	3208
4	Silver	3190

7. Average Age by Gender

	Sex text 	average_age numeric 
1	M	26.26
2	F	23.78

8. Top Events by Participation

	Event text	participants bigint
1	Football Men's Football	1738
2	Hockey Men's Hockey	1133
3	Ice Hockey Men's Ice Hockey	1009
4	Basketball Men's Basketball	921
5	Water Polo Men's Water Polo	911
6	Cycling Men's Road Race, Individual	737
7	Rowing Men's Coxed Eights	648
8	Athletics Men's 4 x 100 metres Relay	591
9	Athletics Men's 100 metres	585
10	Gymnastics Men's Individual All-Arou...	583

9. Countries Participating in Most Sports

	NOC text	sports_count bigint
1	USA	57
2	GBR	57
3	FRA	55
4	CAN	53
5	ITA	52
6	AUS	48
7	GER	48
8	RUS	46
9	JPN	46
10	SUI	45

10. Average Height and Weight by Medal Type

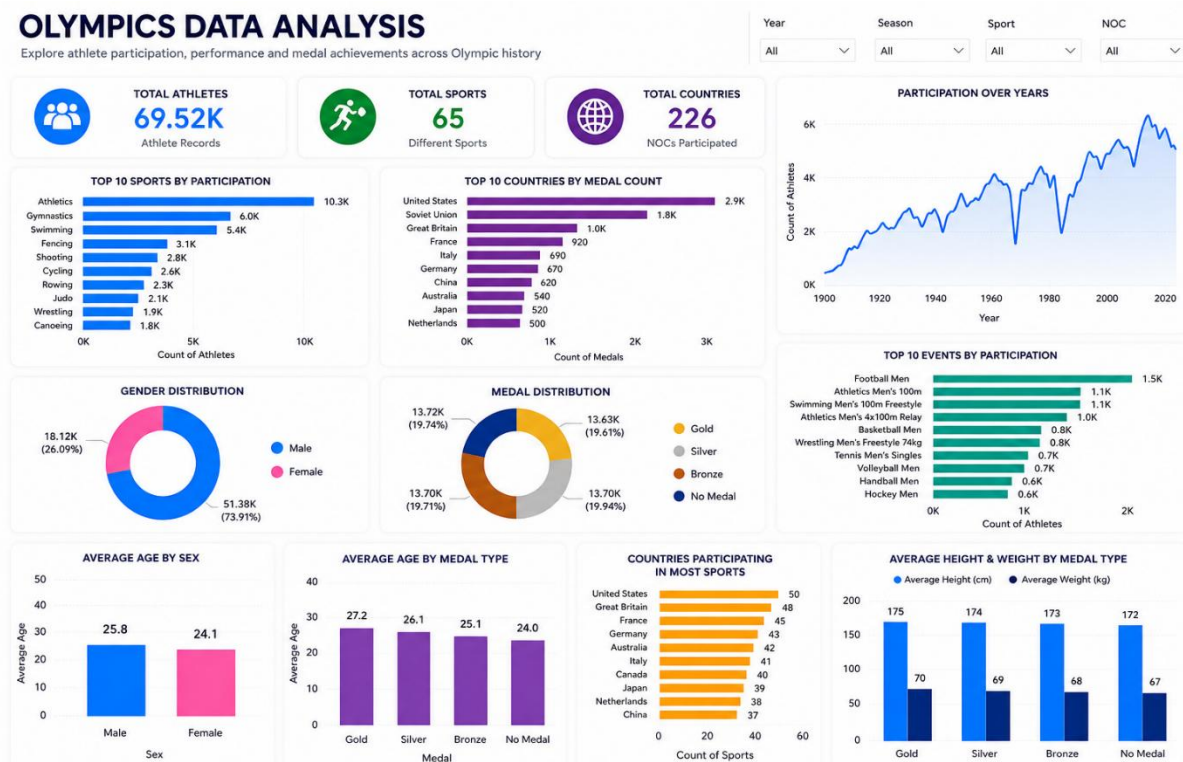
	Medal text 	avg_height numeric 	avg_weight numeric 
1	Bronze	177.04	72.85
2	Silver	177.26	73.12
3	Gold	177.14	72.78

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually

OLYMPICS DATA ANALYSIS

Explore athlete participation, performance and medal achievements across Olympic history



6. Key Insights

- Olympic participation increased significantly over the years.
- Athletics and Swimming consistently attract the largest number of athletes.
- A small number of countries account for a large proportion of Olympic medals.
- Male participation historically exceeded female participation.
- Physical attributes such as height and weight vary across medal categories